

Innovators

“The Future Of AI Computing Can Now Run On A Battery-Powered Device”

What if AI could deliver the accuracy of digital computing while using only a fraction of the power? How close are we to a world where intelligence is seamlessly embedded in everything from wearables to industrial systems without compromise? Ambient Scientific's GP Singh reveals the story behind breakthrough mixed-signal AI processors and the bold vision to redefine the future of computing in conversation with Nidhi and Vidushi from EFY.

GP SINGH
CEO, AMBIENT SCIENTIFIC



What does your company do?

We are a fabless semiconductor company. Production is handled by contract manufacturers. We focus on advanced AI processors, ranging from simple components such as resistors to complex microprocessors that run hundreds of applications, including Intel x86, ARM chips in smartphones, and GPUs.

The next era of computing is AI and deep learning using neural networks. Few companies design microprocessor-level AI chips, and we are one of them. Our core technology, Digan or MX8 Engines, can be scaled like GPUs. Our first product, GPX 10, has 10 AI cores, and GPX 10 Pro also has 10 cores but with more memory and broader capabilities. These chips make it possible to run AI on battery-powered devices, which was previously impractical due to high power requirements.

What was the core idea behind creating an AI-native computer architecture?

Over the past four to five decades, new types of computing

have led to new processors. Mainframes relied on IBM's i390, desktops introduced x86 and Intel, mobile devices adopted ARM, and graphics-intensive workloads required GPUs. Today, AI and deep learning are driving the next shift, requiring processors designed specifically for these workloads.

We recognised this early and began designing AI microprocessors from the ground up. Power consumption was a major challenge, so we developed a low-power AI compute core. This approach improves AI efficiency while directly addressing industry-wide power constraints.

How innovative is your product, if you had to summarise it in a few words?

Analogue AI is a natural fit for AI workloads because approximately ninety-five per cent of AI computing involves matrix operations, known as MAC computing. Analogue architectures handle these operations exceptionally well, while digital computing manages the remaining tasks. Historically, errors

and reliability issues have limited the adoption of analogue AI, but our innovation overcomes these challenges, positioning us to lead in commercial AI computing for the next two decades.

By combining digital and analogue architectures, we enable developers to use existing software tools without sacrificing accuracy. This seemingly small change creates a paradigm shift and defines a clear roadmap for the next two decades.

Can you explain the difference between the AI core and other cores such as CPUs, GPUs, and traditional microprocessors?

Traditional computing with CPUs, GPUs, or microprocessors relies on binary decision trees, where the system makes one decision at a time, with each step dependent on the previous one. In contrast, AI computing is based on pattern matching rather than sequential calculations, using a data-flow architecture instead of a control-flow architecture.

While traditional computers are optimised for a small number